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RESEARCH TOPIC

Investigating French equity portfolio diversification with GSCI in the 21st century

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0 Abstract

In the early 2000s, passive investment in commodity indices has seen a boom in popularity, promising competitive returns and a diversification opportunity for equity portfolio owners. This promise fell short under expectations, as the larger body of investors taking actions in both markets simultaneously contributed to a growing interdependence structure between them, eroding diversification opportunities, this is the financialization era.

We extend existing research on the question of financialization by exploring the specific case of French CAC40, with respect to the traditionally used S&P 500, to represent a diversified equity portfolio. This choice is motivated by the relatively smaller size of the French economy, less inclined to weight on the international commodity markets, and the fact the French economy is mostly a service economy. Doing so, we hope to isolate and capture better the "financialization effect" of returns on the returns on a popular commodity index, S&P GSCI.

Using a Quantile regression, we are capable of providing evidence on the shape of the dependence structure of the returns on S&P GSCI, in particular we observe extreme returns are better explained by financial variables, signaling fading diversification properties in extreme market conditions, regardless of the direction of index price movement.

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1 Introduction

Diversification is a recurrent subject of portfolio and investment theory in general. Starting from the 2000s, the idea of diversifying an equity portfolio with commodity futures has gained popularity, whether in academia or among practitioners, leading to the massive adoption of commodity index investing, notably the S&P GSCI, launched in 1991, and the Dow Jones-UBS Commodity Index (DJ-UBSCI).

As predicted in Barberis and Shleifer 2003, this would eventually lead to an increased interdependence among commodities futures contracts prices. On a broader scale the rebalancing needs and fear contagion from actors now invested in both equity and commodity markets, by means of index investing, would inevitably increase the interdependence between the two markets. Tang and Xiong 2012 provide empirical evidence of this phenomena, and the 2004-2014 is now referred to as the financialization era.

Due to this evolving dependence relationship between equity and commodity futures markets, extensive work has been published to try and understand the matter. Naturally, copulas methods, for their mathematical and elegance and their practicality for explaining joint distributions (a consequence of Sklar's Theorem), attracted a lot of interest, and now constitute an important body of the research on the financialization (Embrechts, McNeil, and Straumann 1999; Wen, Wei, and Huang 2012; Aloui, Hammoudeh, and D. K. Nguyen 2013; Kharoubi-Rakotomalala, Maurer, et al. 2013; Du and He 2015; Gatfaoui 2016). In parallel to this, other methods, like the quantile regression methods, helped understand joint dependence structure by providing linear models predicting a fixed quantile of the distribution of the explained variable given explanatory ones (Baur 2013; Baur, Dimpfl, and Jung 2012; Bianchi, Fan, and Todorova 2020), this is the method we will use for this thesis.

Most of the research on the subject of financialization has been trying to explain movements in S&P 500 (Adams, Collot, and Kartsakli 2020; Gatfaoui 2016; Bianchi, Fan, and Todorova 2020), despite some attempts were made at explaining this phenomena in other parts of the world (Mensi, Hammoudeh, et al. 2021 explores its implications for risk

propagation for EU investors). We propose to study the dependence structure between commodities markets and CAC40. We argue this idea to be original and relevant, as French economy more service oriented, its energy mix singles out by the prominence of nuclear energy, absent of the main commodity indices, and that its aggregate production is not sufficient to impact commodities markets on a fundamental level. Consequently, we argue this makes it a strong candidate to single out the effects of financialization of commodity futures markets, without the noise that fundamental considerations (typically on the underlying state of the economy reflected by the S&P) would have in our models predicting returns on S&P GSCI, which was chosen taking into consideration Tang and Xiong 2012; Bianchi, Fan, and Todorova 2020; Barberis and Shleifer 2003; Barberis, Shleifer, and Wurgler 2005, that most likely it will be more financialized, by the sole fact of its popularity among investors seeking diversification, than individual futures. Furthermore, investigating the dependence structure of the index with the stock market, rather than focusing on different commodities futures, may be much more realistic of the typical investor's behavior, and what dependence structure they may expect.

Before delving into our problem, we will review existing literature in Section 2. More precisely, we will fist recall the theoretical framework of portfolio optimization problems and the consequent attraction for diversification (in particular the attention surrounding correlations). This will serve as a starting point as to understand the growing enthusiasm for commodity indices investing in the early 2000s, marking the beginning of the financialization. Then we will retrace the evolution of the research on financialization, have a brief word on copulas, a prevalent tool to understand this phenomena, and the quantile regression, which will be the base for our empirical studies.

This literature review will be followed by an exploration of our data in Section 3, during which support further the choice of the different indices by providing additional information regarding their composition, as well as a qualitative overview of French economy.

In Section 4, we will present the Quantile Regression method, first introduced by Koenker and Bassett 1978, a Goodness of Fit measure which will serve to understand the quality of our model, which we will interpret as a the explainability of the GSCI returns using financial variables, and the Quantile-Autoregression (QAR, see Koenker and and 2006), which will be the extension we will be using.

A second part of this section is dedicated to the formulation of our model, the various sets of explanatory variables considered, and the potential issues these choices arise.

Section 5 will present our results and compare them with existing research on the subject.

In Section 6 we conclude our study.

2 Literature review

2.1 Portfolio Theory

Before delving on the various impacts of financialization on Portfolio Selection, we provide a quick reminder of elementary concepts of portfolio selection theory.

2.1.1 Theoretical Framework

Definition 2.1.1. *Consider* $(\Omega, \mathcal{F}, \mathbb{P}), (I_t)_{t\in\mathbb{N}}, \mathcal{A}$ *where* $(\Omega, \mathcal{F}, \mathbb{P})$ *is a probability space,* $(I_t)_{t\in\mathbb{N}}$ *is a filtering and* $\mathcal{A}$ *simply denotes the set of available assets, where an element* $S \in \mathcal{A}$ *is understood as a function that assigns to time* t *the random variable* S_t*, and such that* $(S_t)_t$ *is a* $(I_t)_t$*-measureable stochastic process.*

We also define, for $S \in \mathcal{A}$ *the return between time* t *and* $(t+1)$ *as the random variable*

$$R_{S,t} = \frac{S_{t+1} - S_t}{S_t}$$

Finally, we can define the portfolio selection problem between time t *and* $(t+1)$ *as the maximization problem given by*

$$\max_{(w_S)_{S\in\mathcal{A}}} \mathbb{E}\left[U\left(\sum_{S\in\mathcal{A}} w_S R_S\right)\middle| I_t\right] \qquad \text{such that} \qquad \sum_{S\in\mathcal{A}} w_S = 1 \tag{2.1.1}$$

Where $U : \mathbb{R} \to \mathbb{R}$ *is simply the investor's utility function given the financial return of his portfolio, that we generally suppose increasing, and whom convexity reflexes investors behavior towards risk (risk aversion will be characterized by a concave utility curve, risk-neutrality with a flat curve and risk-appetence with a convex one)*

Remark 2.1.1. *We typically cover the case where* $\mathcal{A}$ *is composed of stocks, bonds, commodities and their related futures and forward contracts, in addition to the what is considered a "risk-free" asset* r*, allowing individuals to borrow and land money at a given*

fixed rate, and with the additional constraint $w_S \geq 0$ *for* $S \in \mathcal{A} \setminus \{r\}$*. We can refer to this class of portfolios as long only portfolios.*

2.1.1.1 Markowitz Portfolio Selection

Definition 2.1.2 (Efficient Market Hypothesis)**.** *The model relies on the following assumptions, governing the evolution of the asset price :*

1. *No arbitrage and no transaction costs (in particular, there is no bid-ask spread and no rebalancing costs)*
2. *Stocks do not pay dividends*
3. *Perfect granularity of assets (ie the* w_S *can be taken over the whole of* $[0, 1]$
4. *Constant risk-free rate* r

In Markowitz 1952, they propose the following way of solving the portfolio selection problem: Let, for $w = (w_S)_{S \in \mathcal{A}}$,

$$R(w) := \sum_{S \in \mathcal{A}} w_S R_S \qquad V(w) := Var(R(w)) = Var\left(\sum_{S \in \mathcal{A}} w_S R_S\right) \tag{2.1.2}$$

Assuming $r \notin \mathcal{A}$, a practical "solution" to the problem by finding an "optimal" portfolio for a given maximal level of variance V_{max} (or equivalently a minimal level of returns) can be estimated by

$$\max_{\substack{w=(w_S)_{S \in \mathcal{A}} \\ \hat{V}(w) \leq V_{max}}} \hat{\mathbb{E}}(R(w)) \tag{2.1.3}$$

Where $\hat{\mathbb{E}}$ is the empirical mean, taken as an estimator of $\mathbb{E}$, and similarly $\hat{V}$ is the historical variance taken as an estimator of V, both taken over a specified time-frame.

Then, we can define as $\mathcal{P}$ the set of efficient portfolios, as the set comprising portfolios allocations $(w_S)_{S\in\mathcal{A}}$ that are solution to the maximization problem for a given level of variance, and the efficient portfolio frontier as the curve obtained by plotting for each variance in the x-axis, the estimated return of the "efficient" portfolio (in the sense of Equation 2.1.3) for the maximum level of estimated variance $\hat{V}$.

Finally, when reintroducing the "risk-free" asset to $\mathcal{A}$, the curve now takes the shape of a tangent to the earlier obtained efficiency curve.

2.1.2 Diversification

Observe that, in 2.1.2

$$Var\left(\sum_{S\in\mathcal{A}} w_S R_S\right) = \sum_{S\in\mathcal{A}} w_S^2 Var(R_S) \quad + \quad \sum_{\substack{S_i,S_j\in\mathcal{A}\\ S_i\neq S_j}} w_{S_i} w_{S_j} Cov(S_i, S_j) \tag{2.1.4}$$

In particular, finding assets with low, and even negative covariances will allow one one to achieve reduced portfolio variances, this is the "diversification effect" (see Mao 1970)

Naturally, investors and academia have been highly interested in diversification. In a more general framework than investing in stocks, as individuals have access to other asset classes (bonds, government bonds, commodities, and more) the studying of the relationship between (and within!) various asset classes has become of prevalent importance in order to achieve greater diversification.

In this thesis, we will particularly attach ourselves to present recent advances in understanding the opportunities to diversify an equity portfolio using commodities futures. We will try to understand the effects of the "financialization" of commodity markets, which their increased dependence to equity markets, in a period from around 2004 to 2014 according to most specialist on the matter.

2.2 The Financialization of Commodity Markets

2.2.1 An historical perspective

Gorton and Rouwenhorst 2006 described commodity futures as a "relatively unknown" asset class, and performed an empirical study over data from July of 1959 to December of 2004 to understand different correlations between the commodity futures with more popular classes like stocks and bonds. The article concluded commodity futures were an attractive asset class for investors desiring to diversify traditional portfolios of stocks and bonds.This article and others reflect the growing interest of academia and private investors for commodities in the early 2000s, which will lead to an explosion in institutional investments in commodity futures and commodity indices, with a total value of index–related instruments purchased increasing from $15 billion in 2003 to more than $200 billion in mid-2008 when adding market to OTC activity (CFTC 2008). Geman and Kharoubi 2008 prove that, using copulas to better explain the structure of interdependence between oil and S&P500, that both are negatively linked to each other, over a period of mostly prefinancialization, consisting of quotes from 1990 to 2006.

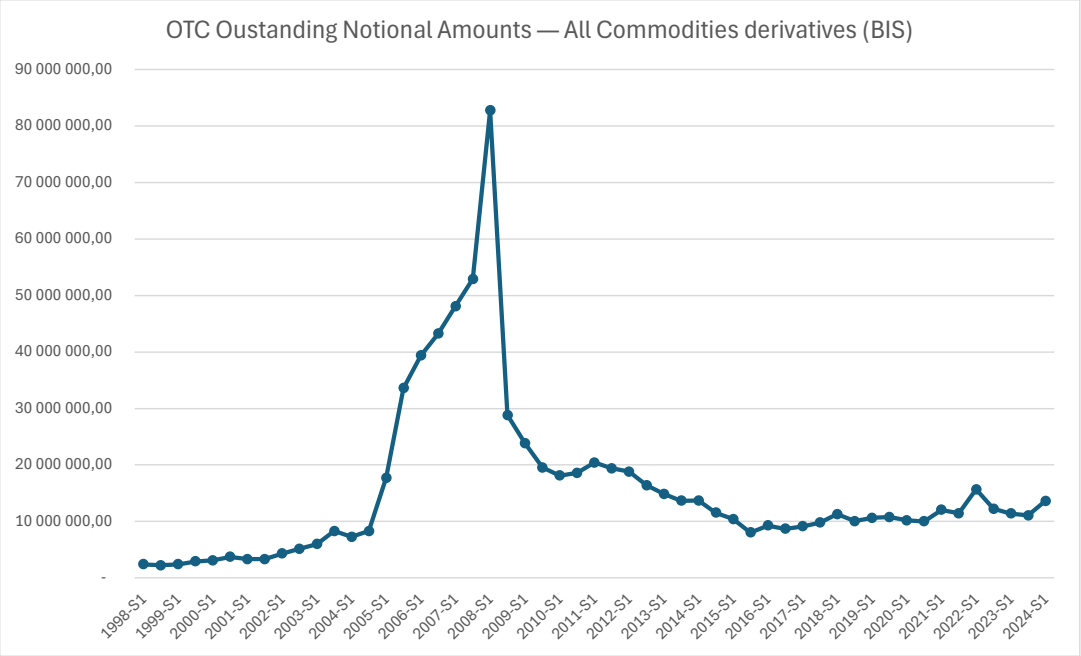


Figure 1: Bank of International Settlements figures of semestrial OTC Commodities Outstanding Notional Amounts at half-year end (end-June and end-December). Amounts denominated in currencies other than the US dollar are converted to US dollars at the exchange rate prevailing on the reference date.

The celebrated Tang and Xiong 2012 proposed one of the first empirical studies of financialization. In particular, they identify index investing in popular commodity indices (namely the S&PGSCI and DJ-UBSCI) from 2004 onward had highly contributed to the empirical evidence of growing comovements between stocks and bonds markets with commodity futures, in particular the ones listed in the aforementioned popular indices. In addition, leaning on the argument from Barberis and Shleifer 2003 that index investing caused increased comovements among basket's components, they evidentiate that commodity indices had been moving more conjointly with oil prices, which were chosen for their study due to the important share they take in major commodity indices. Silvennoinen and Thorp 2013 further details, by means of double smooth transition conditional correlation (DSTCC–GARCH) models, the evolution of correlation between stocks, bonds and commodity futures, from a near zero point 1990, a steep increase in the early 2000s before peaking during the 2008 crisis. Adams, Collot, and Kartsakli 2020 adds that during the financialization period, by means of decomposition of an OLS Regression's R^2 of crude oil returns, financial variables were responsible for 56% of the R^2, against 21% for traditional economic variables, reinforcing the interpretation of crude oil as a financial asset. Furthermore, the authors observe that financial variables do not only have an explanatory power in predicting commodity returns but may also transmit risk to commodity markets. This is coherent with the aforementioned impacts of the increased commodity investing in the early 2000s, as investors in both markets portfolios rebalancing may propagate the risk into commodity markets. Such herd movements propagate volatility across the different markets, as discussed in Information and Volatility spillovers.

2.2.1.1 The Copula Approach

The complex dependence relations between commodities and stocks has been subject to extensive studies. In particular, oil and gas have been scrutinized with particular attention, due to their ties to inflation and monetary policies (see Gagliardone and Gertler 2023). Because of the non-gaussian nature of asset returns, traditional investigations of correlation between time series fall short to understand the true nature of the joint distributions of commodities and stocks. This will lead to the introduction of other econo-

metric tools to investigate dependence structures between assets, notably the copulas (or copulae).

One of the first formulations of copulas, given by Sklar 1973, is the following

Definition 2.2.1. *An n-dimensional copula (briefly, an n-copula) is an n-place real function C with* $\operatorname{Dom} C = I^n$, $\operatorname{Ran} C = I$, *which satisfies the following conditions:*

$$C(1, \ldots, 1, x_m, 1, \ldots, 1) = x_m \quad \textit{for each } m \leq n \textit{ and all } x_m \in I, \tag{7}$$

$$C(x_1, \ldots, x_n) = 0 \quad \textit{if } x_m = 0 \textit{ for any } m \leq n, \tag{8}$$

$$C \textit{ is n-increasing.} \tag{9}$$

Theorem 2.2.1 (Sklar)**.** *For $n \geq 2$, let G be an n-dimensional distribution function with 1-margins $F_1, \ldots, F_n$. Then there exists an n-copula C such that*

$$G(x_1, \ldots, x_n) = C(F_1(x_1), \ldots, F_n(x_n)) \tag{10}$$

for all n-tuples $(x_1, \ldots, x_n)$ in E^n. Conversely, let C be an n-copula and $(F_1, \ldots, F_n)$ an n-tuple of 1-dimensional distribution functions. Let an n-place real function G be defined via (10). Then G is an n-dimensional distribution function with 1-margins $F_1, \ldots, F_n$.

And most importantly, its corollary :

Corollary 2.2.1. *Let $X_1, \ldots, X_n$ be E-valued random variables defined over a common probability space. Denote the individual distributions functions of $X_1, \ldots, X_n$ by $F_1, \ldots, F_n$, respectively, and the joint distribution function of $X_1, \ldots, X_n$ by G. Then F_m is the mth 1-margin of G for all $m \leq n$, and there exists an n-copula C (a connecting copula of $X_1, \ldots, X_n$) such that (10) holds for all $(x_1, \ldots, x_n)$ in E^n. If each F_m is continuous, then the connecting copula C of $X_1, \ldots, X_n$ is unique.*

In order to address intricate questions of distributions, copulas methods have gained popular interest due to

Since the celebrated Embrechts, McNeil, and Straumann 1999, Geman and Kharoubi 2008 were some of the first authors of the growing body of literature applying copulas to understand asset joint distributions, examining the interdependence of WTI Crude Oil futures of different maturities with the stock market. Later, C. C. Nguyen and Bhatti 2012 apply copulas to understand this effect specifically on the Chinese and Vietnamese markets.Wen, Wei, and Huang 2012 introduce a time-varying copula approach to measure contagion between stock market and energy markets during the 2008 crisis. Aloui, Hammoudeh, and D. K. Nguyen 2013 apply a similar time-varying copula approach to the case of Central and Eastern European stock markets, in light of their different energy mixes, despite all being net oil importers and industrialized countries. Gatfaoui 2016 uses a regime-switching model, in addition to copulas, to study the impact of the variance levels in stocks and commodity on their interdependence and particularly the dependencies in the tails of their distributions (hence the use of copulas). She concludes that the relation between US Equities and Oil and Gas are unstable over time, complicating forecasts. Du and He 2015 explore the risk contagions between stock and commodity markets by associating copula methods to an original an original approach, introducing the concept of upside Value at Risk (in addition to the usual "downside" VaR) time series for the different assets, using mixtures of GARCH and GJR-GARCH models. He defines the "upside VaR" as the upper α-quantile of the conditional probability distribution:

$$\text{Upside Var at time t} := \inf\{x \in \mathbb{R} : P(Y_t > x \mid I_{t-1}) \geq \alpha\} \tag{2.2.1}$$

Where $(Y_t)_t$ is the process for which the VaR is estimated, I_t is the information available at time t and α is arbitrary in $[0, 1]$.

All in all, copulas have gained great popularity due to their ability to model complex and nonlinear relationships (as tail dependence or asymmetric dependence), which can be useful to understand the impact of extreme market movements, which is essential for risk testing. In addition, the particular choice of the copulas (among the various families of copulas like the Archimedean, elliptical or non-parametric) and their particular specifications and, when is the case, parametrization, remain as many open problems in academia, as reported in Kharoubi-Rakotomalala, Maurer, et al. 2013.

2.2.1.2 The Quantile Regression Approach

Introduced in Koenker and Bassett 1978, the Quantile Regression estimates, for a given quantile, the conditional quantile of a given variable. It is considered an extension of the typical OLS linear regression, with the advantage to being more robust to the impact of extreme values in the data. The model was originally developed as an alternative to OLS regressions in cases of non-normality of errors, but was later applied by Baur 2013 as an alternative method to copulas to investigate the degree and structure of dependence between a dependent variable and its explanatory variables. More precisely, fitting the quantile regression model for various quantile levels in the dependent variable, he was able to describe how the explanatory power of each regressor evolve, as well as the overall predictive power (via a Pseudo-R^2 metric, see Goodness of Fit). Bianchi, Fan, and Todorova 2020 later use this method to explain the relation between commodity markets and equity markets, investigating the relation between S&P returns and those of different commodities. They point out a strong effect of the financialization on the dependence structure between 2004 and 2013. On the other hand, post 2013 results are more contrasted, with an increased dependence of energy commodities to financial variables, and a decreased one for metals. The structure of dependence of the different commodities is also pointed out to evolve over time, revealing stronger/weaker effects on the tails of the distributions, which they attribute to investor's behaviors in extreme settings, providing evidence of the complex nature of the relation between equities and commodities, and casting criticism on the traditional correlation based approaches. Sharma 2022 extend those results by applying a similar approach over the COVID-19 period, and indicate insignificant effects of stock markets on commodity prices' on upper and lower tails of the quantiles of the distribution. They also investigate equity commodity relationship using other national equity markets, and provide evidence for the stronger dependence structure for markets of more developed countries, casting doubts on diversification opportunities. Xiao et al. 2019 apply a quantile regression to prove that, based on data from March 2011 to May 2018, the impacts of oil implied volatility (CBOE's OVX index) significantly positively impacts the China ETF Volatility index (CBOE's VXFXI). Otherwise, quantile regressions have had applications beyond the sole explanation of financialization: Engle and Manganelli 2004 constructed the Conditional Autoregressive Value at Risk by Regression

Quantiles (CAVR) method for risk-management, while Feng, Chen, and Bassett 2008 provide a new framework to measure momentum based on quantile estimates.

2.2.2 A theoretical framework of Financialization

Economists tried to propose theoretical models to explain the financialization of commodities. In the celebrated Basak and Pavlova 2016 the authors suggest a model with microfoundations with the following, assuming a fixed number K of commodities, a fixed time horizon T and the values of the different commodity are Generalized Wiener Processes, ie of the form

$$dD_t = D_t a\, dt + b\, dW_t \tag{2.2.2}$$

Where a and $b > 0$ are constants. The stocks are modeled in a similar fashion. Two classes of investors are considered. They are either

- Normal, in which case their utility will be a simple concave of the final value of their portfolio.

$$U_{Normal}(\text{Portfolio's Final Value}) = log(\text{Portfolio's Final Value})$$

- Institutional, in which case their utility will also depend on the final value of a given index taking values $(I_t)_{0\leq t\leq T}$ which is simply a linear combination of the values of some of the aforementioned commodity assets available on the market.
  Their utility function can be given by

$$U_{Institutional}(\text{Portfolio's Final Value}) = (\alpha + \beta I_T)log(\text{Portfolio's Final Value})$$

  Where $\alpha, \beta > 0$ are constants, the idea being to capture the effects of benchmarking with respect to I_T : When the benchmark is high the demanded performance is more important so investors have a greater marginal utility in the portfolio's value. Conversely when the benchmark index is low it can be exceeded with lower portfolio values and investors are less incentivized to produce increased portfolio values, leading to lower marginal utilities.

The paper concludes that in presence of institutional investors, shocks in the fundamental demand in index commodities get transmitted to the prices other commodities and that volatilities and correlations of commodities returns increased with institutional presence. These two results are coherent with the empirical higher interdependence among commodities observed earlier. On the other hand, they model also concludes on the propagation of shocks from the stock market to the commodity market. For these reasons this model has had a lot of popularity among the academic community.

2.2.3 Information and Volatility spillovers

At heart of understanding financialization lies the question of understanding how does one course of actions being taken in a given market, or perhaps a news release, propagate to movements in the second one.

Drake et al. 2017 introduce the concept of attention comovements, as a more general question and a possible explanation to asset comovements. This approach is coherent with the earlier claims on investors implied in the commodities and stock markets to being partly responsible for their increased codependency in financialization era, as we could interpret invested amounts as proxies of investors' attention. The article from Drake et al. 2017 supplements this traditional approach by its original formulation of the comovements problem as an "attention comovements" problem, and trying to answer through variables accounting for the attention given by the medias or the number of Google searches for the firm stocks. They argue that comovements in attention is positively associated with returns' comovements of individual stocks, a result in line with the ones from Barberis, Shleifer, and Wurgler 2005. Aromi and Clements 2019 explains how information flows, in the form of media attention and internet searches, impacts the links between oil and equity markets. They reveal the rate of information flow was highly related to the level of oil and equity markets volatilities, but had little impact on their correlation. El Hedi Arouri, Jouini, and D. K. Nguyen 2011 investigate how shocks from oil markets are transmitted to EU and US markets and conversely, from 1998 to 2009. They observe that lagged returns from two sectors in Europe (financial and utilities) and four sectors in the United States (automobile & parts, financials, industrials, and utilities) generate

an impact on oil market. They also provide empirical evidence that volitility spillovers only go from oil to stock sectors in Europe, and go both ways between oil and American Equities.

2.2.4 Implications for portfolio management

As expected, financialization can only be detrimental to the possibilities of diversification of equity portfolios using commodities. Fry-McKibbin and McKinnon 2023 document the initial idiosyncratic nature of commodity returns, and their gradually increasing interdependence to the stock market, driving down diversification opportunities, they observe this phenomena is particularly pronounced for oil, as well as the increased interdependence of commodities among them. This marks a clear shift from prefinancialization articles on the subject (see Gorton and Rouwenhorst 2006; Geman and Kharoubi 2008). As mentioned earlier, Du and He 2015 employ a copula method to evidentiate the extreme risk contagion between the two markets studied, casting doubts on diversification properties when most needed, particularly in bearish markets. On a mostly prefinancilization era dataset (1970-2005) Cheung and Miu 2010 advocate diversification existed in the long run, due to infrequent commodity outbursts (one may think of the recent outbursts of nickel prices and cocoa futures). Nonetheless, they argue diversification properties of commodities for portfolios of US or Canadian stocks fade when these later are bearish, hence when diversification is most needed, and conclude diversification properties of commodities are less observable for stocks from resource-based economies like Canada. These results are of prevalent importance, indicating that diversification properties may be more observable in relatively less resource-based economies, so one may think of countries with more important service sectors.

3 DATA

3.1 Data Exploration

The data used includes daily values of CAC40 ranging from December 2003 (earliest availability from Yahoo Finance) up to March 2025, as a proxy for diversified equity portfolios for French investors. As there is no available implied volatility index for CAC40 on yfinance, we use VIX as a proxy, in light of volatility contagion phenomena. The choice of tracking the S&P GSCI is coherent with its great popularity among investors looking to diversify their portfolio with some exposure to commodities, known to be one of the main drivers of the financialization (Tang and Xiong 2012). This index is highly weighted towards Energy Commodities (above 50%), with in particular more than 20% in WTI Crude Oil and in Brent Crude Oil, respectively. This very high presence in this index favors the prominence of Energy Commodities comovements with Equity Indices, when compared to relatively less represented commodity items (Tang and Xiong 2012; Adams and Glück 2015; Bianchi, Fan, and Todorova 2020.

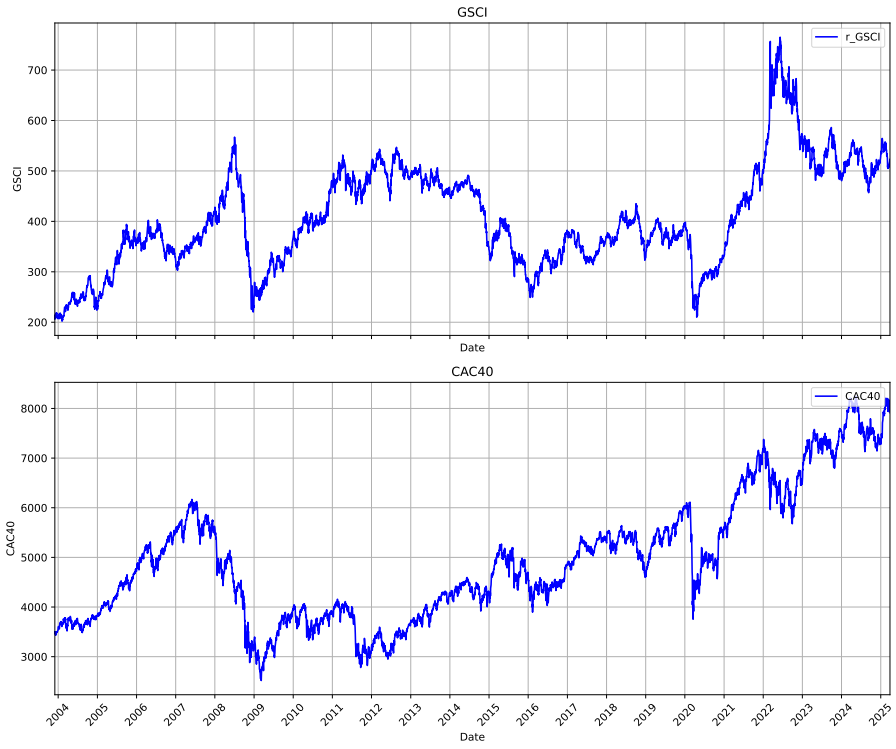


Figure 2: CAC40 and S&P GSCI over time

Table 1: Descriptive Statistics and Stationarity Tests

	Mean	Median	Std Deviation	Kurtosis	Skewness	ADF Test Statistic	ADF p-value	Jarque-Bera Test Statistic	Jarque-Bera p-value
r_GSCI	0.074098	0.147414	3.903356	5.569438	-0.154723	-43.672904	0.000000	6827.017412	0.000000
r_CAC40	0.060968	0.151409	3.317471	8.449895	-0.072896	-13.456599	0.000000	15671.173045	0.000000
vol_CAC_1M	10.062351	8.527386	5.677309	9.712951	2.566340	-6.320300	0.000000	26480.480776	0.000000
vol_GSCI_1M	12.378801	11.326823	5.578612	8.997274	2.441225	-6.414686	0.000000	22992.508079	0.000000
VIX	18.974425	16.469999	8.597785	10.047291	2.608155	-5.531443	0.000002	28119.990746	0.000000

Note: Descriptive statistics, volatilities as reported in % points.

As shown in the Table 1, GSCI and CAC40 returns are left skewed, possibly indicating a fat left tail, and leptokurtic (kurtosis ≥ 3), signaling a non-normality (in the sense au Gauss) of returns, which is later confirmed by the the Jarque-Bera test, significant in both case at the 1% level. Also tested our data for the presence of a unit-root using an Augmented Dickey Fuller (ADF) test. As expected, we reject the null hypothesis there's is a unit root at 1% confidence levels. Figure 3 displays the distribution of returns of both indices under study. The leptokurtic nature is evident, and a slight left-skewness is observable.

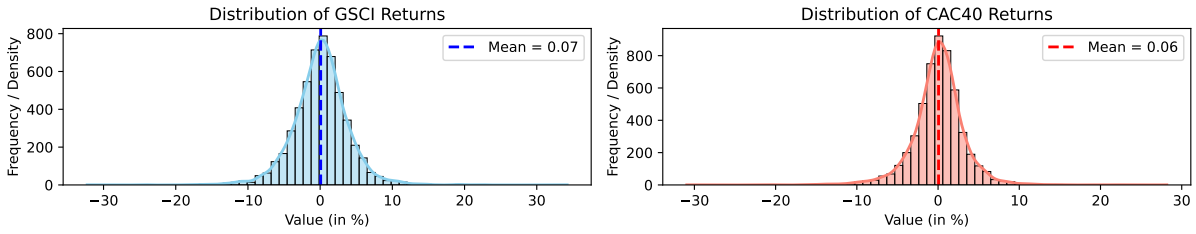


Figure 3: Distribution of GSCI and CAC40 daily returns, 2003-2025

Ahead of our future works using quantile regressions, we introduce Kendall's Tau, which can be interpreted as a correlation on the ranks on the different series, rather than on the series themselves. In particular, it is known to be non-parametric, and better suited for non-gaussian data than the usual (Pearson's) correlation. Figure 4 displays Kendall and Pearson correlations for the three main variables under study. We highlight that Kendall's correlation between CAC40 returns and GSCI returns is significant at the 1% significance level, while the one between VIX and GSCI is not, at 10% level.

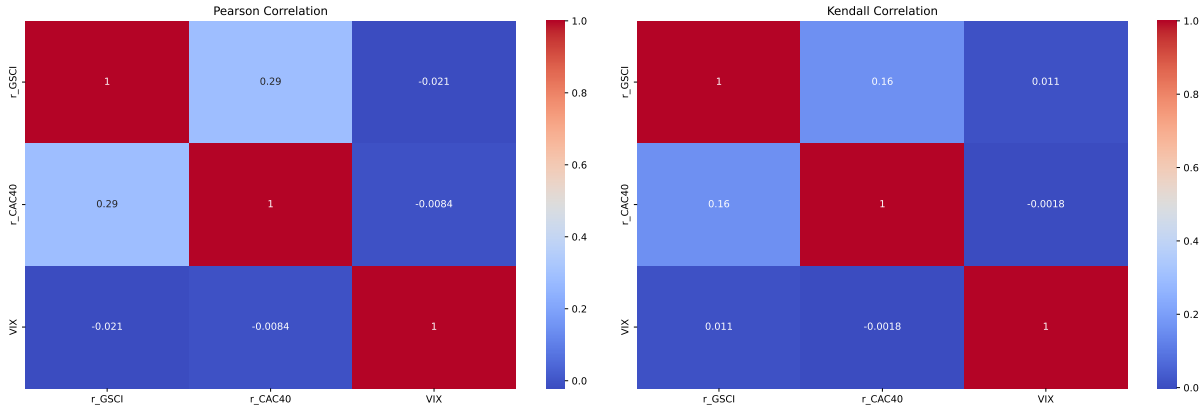


Figure 4: Pearson and Kendall Correlations among main variables of interests

3.2 Why France ? A tour on French economic data

In 2023, with a Gross Domestic Product of 3 051 Billions of USD, France accounted for nearly 3.77% of World economic output (The World Bank 2022). This share of world output, considerably lower than their American counterpart, with a GDP of nearly 27.720 Billion USD in 2023, 26.1% of World GDP, motivates our decision to choose CAC40, and not the usual S&P500 as to understand the impact of financialization on the dependence structure between equity and commodity prices. By being a large producer and demander of various commodities (including oil), the USA's economy is a much more important driver of commodity prices trends over the world, which could result into additional ties between S&P500 and commodity indices than CAC40 would have. Also, we believe that using CAC40 could provide a picture of what dependence structure between national equity market and commodity prices would look like for many citizens of developed, non-commodity driven industries, that may rely more on the service sector. In particular, France, only relies on oil and gas for 42.8% of its energy mix (International Energy Agency 2024), are much less dependent on energy commodities than the USA, with 71.9%. Considering oil and gas are some of the most financialized commodities, and account for more than 57% of the S&P GSCI.

We conclude this section with a couple words on the CAC40 and the S&P GSCI indices.

The CAC40 is a market capitalisation weighted index tracking the 40 largest publicly traded companies on Euronext Paris (*Euronext* 2025), and is often considered a proxy

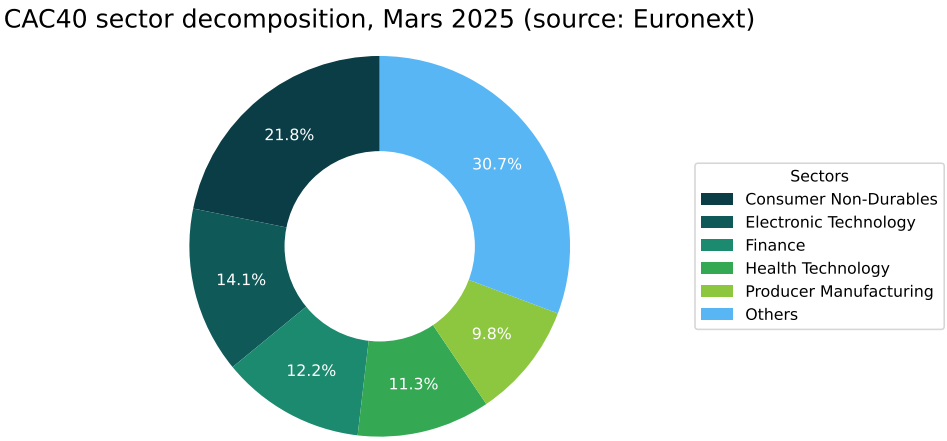


Figure 5: CAC40 Decomposition by sector (source: *Euronext* 2025)

for a diversified porfolio in french equities. The total capitalization of these 40 companies amounted to nearly 2100 billion Euros in 2024 (*Euronext* 2025). We observe from Figure 5 that the index is balanced among various industries, and that the share of companies relying heavily on commodities listed in the GSCI is limited with respect to other national stock indices. On the other hand, we observe that the index is nonetheless highly concentrated among its top 10 stocks, contributing to 59.74% of the index. As per data from 2022 ownership of CAC40 is divided among asset managers ($\approx 26\%$), from which the top five asset managers (of which Blackrock, Vanguard and Capital Group) own nearly 8.6%. Then, families and founders are the second most important owners, controlling 21% of CAC40, followed by individual investors with 5.3% of ownership (Euronext 2024).

Table 2: Top French Companies by Weight in Sector Allocation in 2024

Company	**Mnemo**	**Country**	**Sector (ICB)**	**Weight (%)**
LVMH	MC	France	Consumer Non-Durables	8.28%
SCHNEIDER ELECTRIC	SU	France	Producer Manufacturing	7.33%
TOTALENERGIES	TTE	France	Energy Minerals	7.31%
SANOFI	SAN	France	Health Technology	6.64%
AIR LIQUIDE	AI	France	Process Industries	5.64%
AIRBUS	AIR	France	Electronic Technology	5.44%
SAFRAN	SAF	France	Electronic Technology	5.00%
BNP PARIBAS ACT.A	BNP	France	Finance	4.71%
ESSILORLUXOTTICA	EL	France	Health Technology	4.71%
L'OREAL	OR	France	Consumer Non-Durables	4.67%
			Total	**59.74%**

Note: This table reports the top 10 French companies by weight in sector allocation in 2024 (source: *Euronext* 2025).

Table 3: Owners of CAC40

Owner	**Position (€Bn)**	**Share in 2022 (%)**
Asset managers	548	26.0
Family and founders	442	21.0
Individual investors	111	5.3
Employees	62	2.9
Industrial investors	57	2.7
Foreign States and sovereign funds	54	2.6
French State	46	2.2
Financial institutions	29	1.4
Treasury shares	28	1.3
Pension funds (active management)	15	0.7
Others	2	0.1
Total (Identified)	**1394**	**66.0**
Unknown	712	34.0

Note: This table reports the top 10 owners of CAC40 in 2024 (source: Euronext 2024).

3.3 A quick overview of the S&P GSCI

Introduced by Goldman Sachs in 1991, the GSCI lunched the era of investable commodity indices, and rapidly stood as a reference among investors. The index, since acquired and administered by Standard & Poor's, is composed of futures trading on various markets (NYMEX, ICE, CME, LME and COMEX), and comprises of nearly 57% allocated in Energies, 11.8% in grains, 4.8% in Softs, 9.3% in Livestock and 16.6% in Metals (see Table 4). The index composition is determined quarterly at announced dates, and futures are rolled between the 5th and 9th business days, at a rate of 20% per day, of each month, including rebalancing months. Nonetheless, the administration of the index keeps open the possibility to roll the contract at other dates, in case a market disruption event occurs (see S&P Dow Jones Indices 2025).

In a promotional article of S&P GSCI appearing on Standard and Poor's website (S&P Dow Jones Indices 2024), investment in the index is praised as a safe haven in times of inflation and aggressive rate cuts from central banks. The article further promotes the index, after reminding that, in light of Markowitz 1952, items with low to negative correlations improved portfolio performance, and hence advocate for embracing commodity investing, implicitly through the S&P GSCI, in light of the low correlation between stocks and commodities. In addition to that, they qualify their index as a "True Commodities Market Measure", a view that can largely be challenged by evidence provided by the important bias the index gives to energy commodities (nearly 57% of the total index value in 4) but most importantly by the empirical evidence provided by Tang and Xiong 2012; Bianchi, Fan, and Todorova 2020 that traded commodities included to the index were more tied to the equity markets than the other ones, casting doubts on the extent to which this index is representative of commodities futures markets as a whole and not of a selected group of commodities used by passive investors to try to diversify their portfolios.

Table 4: S&P GSCI 2025 Reference Percentage Dollar Weights by Commodity, Exchange, and Category

Commodity	Exchange	2025 RPDW
Energies		
WTI Crude Oil	NYMEX	20.257120%
Heating Oil	NYMEX	4.223675%
RBOB Gasoline	NYMEX	3.901648%
Brent Crude Oil	ICE	20.726640%
Gas Oil	ICE	5.908821%
Natural Gas	NYMEX	2.393480%
Grains		
Chicago Wheat	CME (CBOT)	2.501533%
Kansas Wheat	CME (KCBT)	1.258336%
Corn	CME (CBOT)	4.558413%
Soybeans	CME (CBOT)	3.506659%
Softs		
Coffee	ICE	1.035835%
Sugar	ICE	1.987869%
Cocoa	ICE	0.819801%
Cotton	ICE	1.026018%
Livestock		
Lean Hogs	CME	2.473356%
Live Cattle	CME	4.672884%
Feeder Cattle	CME	2.156991%
Metals		
Aluminum	LME	3.762849%
Copper	LME	4.954048%
Nickel	LME	0.902610%
Lead	LME	0.570813%
Zinc	LME	0.815317%
Gold	COMEX	5.104469%
Silver	COMEX	0.480814%

4 Methodology

4.1 The Quantile Regression

4.1.1 Model formulation

Let Y be our defendant variable, with cdf F, $F(t) := \mathbb{P}(Y \leq t)$, we define the τth quantile as its image by pseudo-inverse of F:

$$q_Y(\tau) = inf\{t : F(t) \geq \tau\}$$

We define the loss function as $\rho_\tau(m) = m(\tau - \mathbf{1}_{(-\infty,0)}(m))$, where $\mathbf{1}_{(-\infty,0)}$ is simply an indicator function. The predicted values of the model are determined by minimizing the loss function.

In quantile regression, we assume that the conditional τ-quantile of the response variable can be expressed as a linear combination of the predictors:

$$Q_Y(\tau|X) = X^T\beta_\tau.$$

Where X^T is simply the transpose of our vector of dependent variables X

To determine β_τ, we minimize the expected value of the quantile loss function, which leads to the following optimization problem:

$$\beta_\tau = \arg\min_{\beta\in\mathbb{R}^k} \mathbb{E}\left(\rho_\tau(Y - X^T\beta)\right).$$

Therefore, for the sample data we produce an estimate using

$$\hat{\beta}_\tau = \arg\min_{\beta\in\mathbb{R}^k} \sum_{i=1}^{n} \rho_\tau(Y_i - X_i^T\beta).$$

4.1.2 Goodness of Fit

Koenker and Machado 1999 develop a goodness-of-fit statistic we can use to evaluate our model, a specific pseudo-R^2 designed for the quantile regression.

The loss function from earlier can be extended to a function of a vector $u \in \mathbb{R}^d$ by

$$\rho_\tau(u) = u \cdot (\tau - \mathbf{1}_{\{u<0\}})$$

Where, for $u = (u_1, .., u_d)$, $\mathbf{1}_{\{u<0\}} := (\mathbf{1}_{\{u_1<0\}}, ..., \mathbf{1}_{\{u_d<0\}}) \in \{0, 1\}^d$

The pseudo-R^2 is computed as:

$$\text{pseudo-}R^2 = 1 - \frac{\sum_{i=1}^n \rho_\tau(y_i - \hat{y}_i)}{\sum_{i=1}^n \rho_\tau(y_i - \tilde{y})}$$

where:

- $\hat{y}_i$ are the predicted values from the quantile regression model,
- $\tilde{y}$ is the predicted value from a model with only an intercept (i.e., the empirical τ-quantile of y).

4.1.3 Application of the quantile regression

A visual example can be given in the following way : we depart from the celebrated CAPM equation (Markowitz 1952)

$$\mathbb{E}[R_i] = R_f + \beta_i \, (\mathbb{E}[R_m] - R_f)$$

to have

$$\mathbb{E}[R_i] - R_f = \beta_i \, (\mathbb{E}[R_m] - R_f)$$

So that we can estimate the coefficient β_i without assuming a fixed rate, and we deploy

a quantile regression approach to plot risk-premia on Societe Generale's returns against market risk-premia. The risk-free rate here is assumed to the ESTR rate.

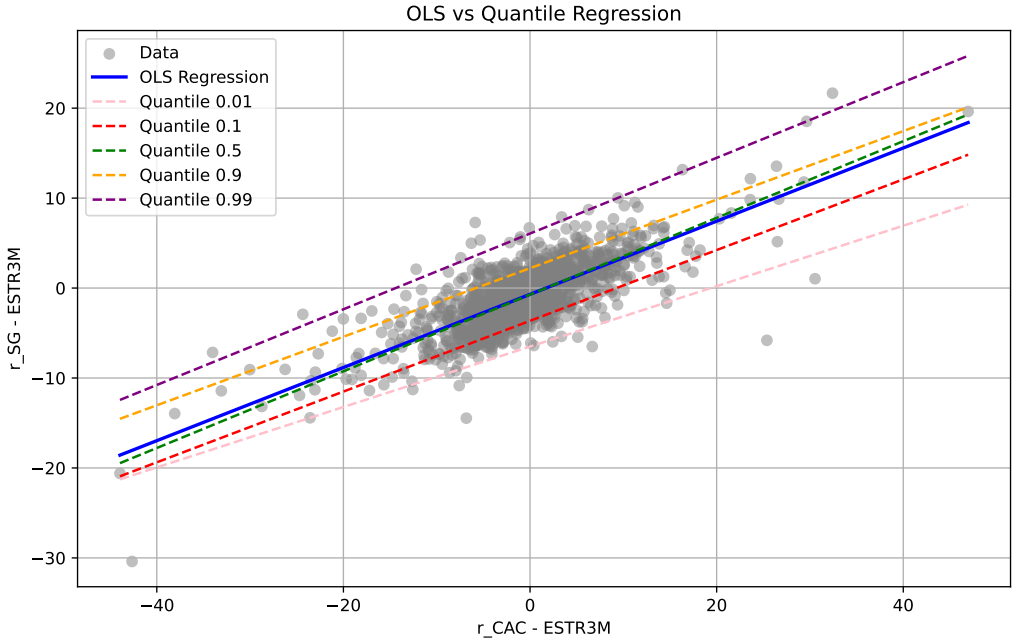


Figure 6: We can see, with the different regression lines, where for each quantile τ the predicted value is expected to be at the τth quantile level

4.1.4 The Quantile Autoregression (QAR) Extension

Koenker and and 2006 propose an extension of their celebrated Quantile Regression, the Quantile Autoregression, to deal with data that has autocorrelation, and, as they argue, help understanding asymmetric dynamics.

$$Q_{Y_t}(\tau|X) = \alpha_0(\tau) + \alpha_1(\tau)Y_{t-1} + X^T\beta_\tau \tag{4.1.1}$$

Where X is a vector of explanatory variables different from Y_{t-1}

4.2 Model Specification

For this thesis, all financial data are imported on Python using Yahoo Finance's API. Python Jupiter Notebooks offer a vast range of data analysis tools and can be shared easily.

Our approach follows the one of Bianchi, Fan, and Todorova 2020, applying a quantile autoregression of financial variables in order to to explain the changes in the S&P GSCI.

In particular, we use interaction variables to adjust for the r_{CAC}'s effects in different time periods, by multiplying to dummies corresponding to:

1. Financialization (January 1, 2004 – December 31, 2013)
2. 2008 Financial Crisis (January 1, 2008 – December 31, 2009)
3. De-financialization (January 1, 2014 – June 30, 2017)
4. COVID-19 Crisis (January 30, 2020 – June 1, 2022)
5. Ukraine War (since December 17, 2021, following Putin's ultimatum to NATO)

Remark 4.2.1. *These five interactions variables are denoted by $\gamma_1, .., \gamma_5$ respectively, in 5*

Observe the first free variables correspond to the ones in Bianchi, Fan, and Todorova 2020, while the two later ones reflect more recent geopolitical events.

Our model is given by, for a quantile level τ

$$Q_{r_t}(\tau \mid \mathbf{X}) = \alpha_0(\tau) + \alpha_1(\tau) r_{t-1} + \beta(\tau) r_{EM,t} + \sum_{j=1}^{5} \gamma_j(\tau) D_j r_{EM,t} + \delta(\tau) VIX_t \quad (4.2.1)$$

Where, for each τ

- r_t is the return on the GSCI index

- $\alpha_0(\tau)$ is a constant
- r_{t-1} is an autoregressed term, as in Baur, Dimpfl, and Jung 2012
- D_j is, for $j = 1, .., 5$ a dummy corresponding to one of the 5 aforementioned periods
- $r_{EM,t}$ is the equity market return at time t (here, taken to be the return of CAC40)
- VIX_t is the level of VIX at time t

The model is calibrated by choosing, for each τ, the coefficients $(\alpha_i(\tau))_{i=0,1}$ and $(\gamma_j(\tau))_{j=1,..,5}$ that minimize the loss function.

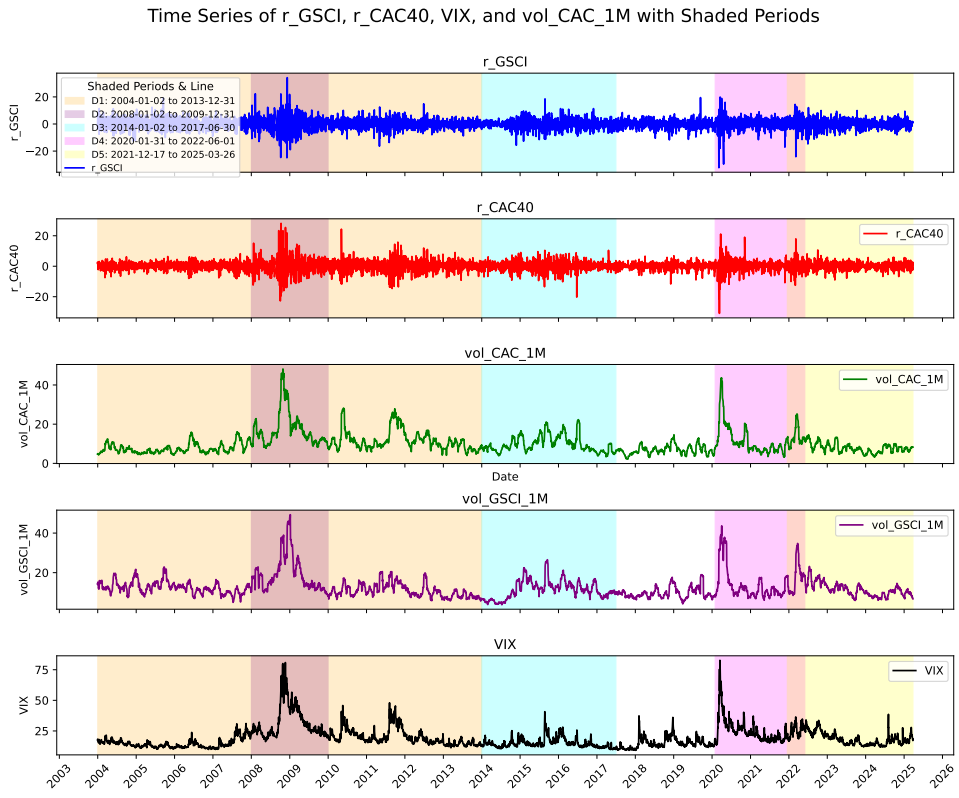


Figure 7: Daily returns of the GSCI, CAC40 and one-month rolling sample variance of returns on CAC40, expressed in % points. Highlighted segments correspond to the regime switches from Bianchi, Fan, and Todorova 2020. It is immediate to observe the joint spikes in volatility, in the highlighted areas.

In Figure 7 we highlight the various time-dummies applied to our study. In particular, we can see similarities in the volatility of both assets, in adequation with volatility spillovers literature between equity and commodity markets (see Mensi, Hammoudeh, et al. 2021; Mensi, Shafiullah, et al. 2021)

5 Results

First, the coefficients of financial variables' r_CAC40 and VIX are significant at the 1% significance level the extreme majority of quantile levels, implying financial variables still have an impacts on commodity markets.

Table 5: Quantile Regression Estimates with OLS Benchmark

	0.05	0.10	0.25	0.50	0.75	0.90	0.95	OLS
Intercept	-3.118***	-2.168***	-0.842***	0.263**	1.095***	2.460***	2.709***	0.231*
$r_{\text{GSCI, lag}}$	-0.113***	-0.084***	-0.081***	-0.049***	-0.080***	-0.054**	-0.025	-0.084***
r_{CAC40}	0.766***	0.563***	0.517***	0.462***	0.339***	0.342***	0.352**	0.578***
VIX	-0.144***	-0.109***	-0.062***	-0.009*	0.054***	0.094***	0.168***	-0.009*
γ_1	-0.535***	-0.242*	-0.162**	-0.123**	-0.030	-0.113	-0.066	-0.262***
γ_2	0.088	0.050	0.052	0.114***	0.108**	0.085	-0.152*	0.062
γ_3	-0.444**	-0.322**	-0.252***	-0.265***	-0.116	-0.025	-0.001	-0.299***
γ_4	-0.020	0.033	-0.021	-0.023	0.048	-0.035	-0.123	-0.111*
γ_5	-0.596***	-0.347**	-0.331***	-0.324***	-0.174**	-0.225*	-0.466**	-0.476***

Note: Quantile regression estimates are reported at the 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles, with OLS results in the final column.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Then, interestingly, we observe different results than Bianchi, Fan, and Todorova 2020 regarding the significance of a lagged term in the model. On one hand, Bianchi, Fan, and Todorova 2020 had mixed conclusions on the significance of a lagged term, with significance disappearing when varying quantiles, and without any clear structure observed on individual (typically the quantiles at which the lagged term would be significant would highly vary between the different commodity items). On the other, our model, which considers the GSCI Index as a whole, evidentiates the significance of the lagged term at the 1% level for all quantiles below 90%, with only a 5% significance at the 90th and no significance at the 95th. This result really encourages the approach that is to consider the GSCI as a financial item per se, and rises questions, in light of the documented increase of pairwise correlations of returns among listed items of the index (see Barberis and Shleifer 2003), on how does the formation of this aggregated unit root, at the scale of the index,

translate to the relation between its various components. Hence, it appears natural to believe this result to be due to the participation to the index itself rather than properties of its underlying commodities futures.

The study of the impacts of the various interaction variables in 5 are only partially echoing those from Bianchi, Fan, and Todorova 2020:

- The coefficient on CAC40 returns being positive, while those of S&P being negative for most commodities in Bianchi, Fan, and Todorova 2020, is most likely due to our dataset not comprising pre-financialization-era data, which would have naturally lowered the coefficient associated with the returns on CAC40 (positive in our study).

- Second and most importantly, our interaction effects for the financialization period are negative. This could be a major setback to our results when comparing to existing theory, but this may only be a spurious effect of our coefficients on CAC40 being too high, and therefore the financialization era (γ_1) is being lowered. In particular, our lack of pre-financialization data translates in the collinearity issue since $r_{CAC40} \approx \gamma_1 + \gamma_3$, resulting in the observed effects. This problem is addressed with the regression in 6.

- However, the interaction terms for the financialization and post-financialization periods remain relevant at the 1% significance level for all quantiles (γ_1, γ_3). The COVID interaction term is highly significant for lower quantiles (1% significance level for quantiles below 90% and 5% significance for the 90th quantile). The interaction effect of the returns on CAC40 during the 2008 crisis (interaction term γ_2) has no statistically explanatory power for extreme returns, being relevant at the 1% significance level for the 50th and 75th percentiles, at the 10% significance level for the 95th percentile regression, and non-significant otherwise. This is coherent with Bianchi, Fan, and Todorova 2020, which observed strong significance of this interaction term only for metals, slight significance for the energies, fading on the majority of the quantiles, and almost non-existent significance on Grains, Softs and Livestock, which sum to about 25% of the whole index, as shown in S&P GSCI 2025 Reference Percentage Dollar Weights by Commodity, Exchange, and Category.

A deeper study of the regression coefficients, in particular the ones on returns of CAC40 and VIX, shows their different impacts on the returns on the GSCI (see 8), for different levels of returns. From this, it seems CAC40 affects less and less the returns on GSCI at higher quantile levels, while we observe the opposite phenomenon for the VIX.

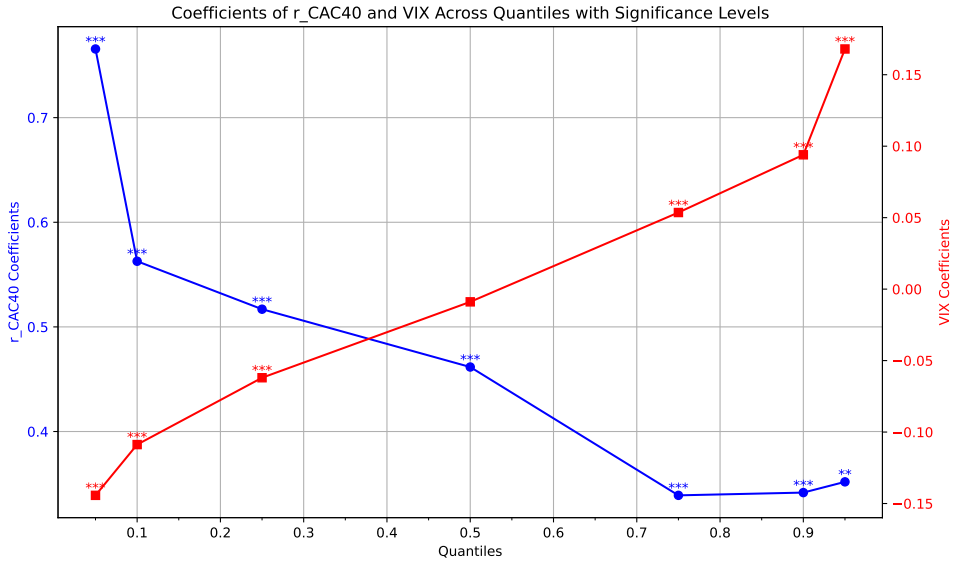


Figure 8: On the x-axis we have the various quantiles for which Quantile Regressions were performed. On the y-axes we plot the coefficients obtained for the regressors of r_{CAC40} and VIX, and display the significance level of the regressors in each model (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$).

Our analysis of the Pseudo-R^2 values signals overall low levels of fit of our model, with a maximum of 13% reached at the 5th quantile, a U-shaped curve with a minimal of 5% at the median and 60th quantile, are consistent with the accepted view commodities futures and indices offer diversification opportunities. Yet, inferences on the diversification properties should due to a small Pseudo-R^2 are to be nuanced by the known non-linear nature of the dependence between the two assets, and their non-normal distributions, which are as many obstacles to the applicability of linear models like the one we use. Otherwise, observe our predictive results are slightly higher overall than those from Baur 2013 when using similar methods to study the financialization of Gold, another popular asset among investors seeking diversification, and which could be either attributable to, in addition of metals, of the presence of energy futures (in particular oil) in our index. Other reasons could include that gold, taken individually, may be more subject to its own fundamental asset price drivers, which diminish the propensity of financial variables to explain the whole of its price variations, while on the other hand, when taken as a whole, the S&P

GSCI receives significant investments for diversification, causing sufficient comovements with the financial markets for reasons of speculation and portfolio rebalancing, as per Basak and Pavlova 2016.

A further investigation of the Pseudo-R^2 values, when plotting its values against the quantiles, allows us to give an approximate picture of the dependence structure between our index's returns and the CAC40 returns, using an approach similar to that of Baur 2013. In particular, the U-shaped curve displayed when plotting the Pseudo-R^2 against the quantiles in Figure 9 means the model performs better on high conditional variations in the GSCI, a result also observe in Baur 2013 for gold futures. This may imply the GSCI index loses some of its diversification properties when large movements occur in the markets, i.e., when diversification is most needed. Also, the curve is non-symmetrical, with Pseudo-R^2 values remarkably higher for below median quantiles, and increasing at the much greater rate the more we deviate from the median, than for their over-median quantile counterparts. This observation signals that the two assets more much more conjointly in bearish market, when diversification is most needed for long investors.

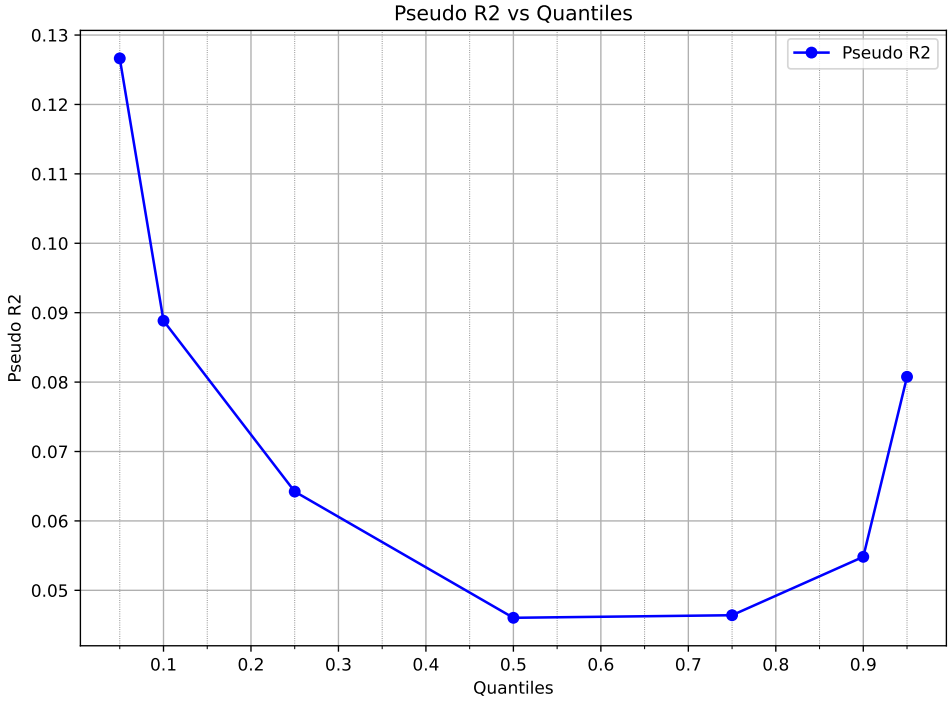


Figure 9: Pseudo-R^2 values across different quantiles. The plot shows how the goodness-of-fit of the quantile regressions varies along the conditional distribution of the dependent variable.

When removing interaction variables γ_3 and γ_5, in order to limit collinearity issues (we just keep financialization, 2008 crisis and COVID interaction variables), we observe the desired results of positive γ_1 coefficients at most quantile levels, which matches Bianchi, Fan, and Todorova 2020's findings. Similarly, the γ_4 coefficients, which were mostly negative in the earlier quantile regressions, are now positive. Nonetheless and most notably, we lose significance in most interaction effects, a setback to our study. Nonetheless, we observe a similar structure on VIX and r_{CAC40} coefficients as in the earlier regression, with the r_{CAC40} increasing as the VIX regressors decrease. The intercepts are at similar levels as earlier, and all variables but the interaction ones remained statistically significant predictors. This result signals most likely the importance of prefinancialization data to be able to mark the switch in investor behavior starting near 2004, which are consistent with conclusions from Bianchi, Fan, and Todorova 2020 that financialization may be more permanent than previously thought, which is also coherent with the strong different in OTC outstanding notional amounts in commodity derivatives between the 2014-2025 years and the pre-2004 years (see Figure 1), which appears to be predicted by the conclusions from Tang and Xiong 2012; Basak and Pavlova 2016.

Table 6: Quantile and OLS Regression Coefficients with Significance Levels

	0.05	0.10	0.25	0.50	0.75	0.90	0.95	OLS
Intercept	-3.143***	-2.131***	-0.839***	0.286**	1.180***	2.442***	2.824***	0.234*
r_GSCI_lag	-0.118***	-0.087***	-0.082***	-0.055***	-0.081***	-0.053**	-0.041	-0.084***
r_CAC40	0.349***	0.282***	0.303***	0.252***	0.221***	0.255***	0.309***	0.288***
VIX	-0.144***	-0.111***	-0.062***	-0.009	0.051***	0.095***	0.165***	-0.009
gamma1	-0.120	0.037	0.051	0.086**	0.089*	-0.025	-0.010	0.027
gamma2	0.101	0.050	0.052	0.110**	0.108**	0.085	-0.179**	0.062
gamma4	0.266*	0.182*	0.159**	0.119**	0.125**	-0.081	-0.218*	0.084

Notes: This table reports coefficients from quantile regressions at various quantiles and OLS. Significance levels are denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

6 Conclusion

After recalling existing research on financialization and quantile regressions, we extended prior works on the question of financialization. That is, our purpose was to examine the non-linear nature of the relation between equity returns of a non-commodity-heavy economy like the French one (characterized by the CAC40 index) with a popular commodity index like the S&P GSCI. The motivation for these choices was twofold: considering the CAC40 index was arguably a better choice to remove the impacts a large economy like the American one on commodity prices (i.e., removing fundamental demand impact), in order to keep only the financialization effect (due to portfolio rebalances); and on the other hand, the choice of the GSCI as a whole justifies itself by the more "financial-like" behavior of the index, with respect to individual commodities futures, and is also more representative of the behavior of a non-expert investor trying to diversify their portfolio by ways of a commodity ETF.

The relationship between the two indices was complex, with interaction effects (CAC40 multiplied by a period dummy) losing and gaining significance when varying time-period and quantile levels, also strengthening existing literature on the complex, evolving nature of commodity-equity dependence structures since the 2000s. First, we observe the model remains consistent with existing evidence from academia despite replacing S&P 500 by CAC40 as the proxy for diversified equity portfolio. We believe similar approaches can be encouraged to isolate the effect of financialization. Mixed results regarding the weights of interaction variables (i.e., changed weights of the CAC40 returns in the model for specific market conditions) as they proved partially inconsistent with some of Bianchi, Fan, and Todorova 2020's weights, suggesting their results could be attributed to the use of pre-2003 data, as they use data since 1990, directing us towards the conclusion that financialization effects are permanent to the extent that, despite being lower than in the 2004–2014 period, they may not be coming back to pre-2004 levels, most likely due to a permanent change in investor behavior and appetite for commodities futures. Importantly, we manage to extend on their result by applying a similar model to the GSCI itself and not to individual futures. An interesting result is the significance of the autoregressive term in the model, despite it not being the case for a majority of

the considered commodities in their article, encouraging us to explore what mechanisms allow for this to happen, in particular how does that translate in the inter-commodity comovements among items of the index.

The analysis of the goodness-of-fit measure (our Pseudo-R^2) highlights two things. First, the overall levels of fit are low, with a maximum of 0.13 reached on the lowest quantile levels, advocating for the standard adage that commodities futures, and in particular passive commodity investing, is an easily accessible way to diversify the risks from an equity portfolio. The reality is more complex unfortunately, as non-normality of returns and non-linear dependence between the considered assets forces us to take some distance to these metrics, in particular it is more interesting to consider their relative levels at different quantiles for a given model, than their values themselves. Consequently, a further investigation displays that our Pseudo-R^2 forms a U-shape when plotting it against quantiles (see Figure 9), supporting the idea that equity and popular commodity indices like the GSCI more strongly depend on one another in times of extreme returns, whether positive or negative, making this diversification approach relatively deficient in market turmoil with respect to the normal market setting. In addition, the dependence structure is more pronounced on below-median returns, and more so as we deviate from the median. This concludes on the relatively week diversification opportunity of S&P GSCI the more we deviate from a "normal" setting of returns, in particular during bearish market, and in light of the complex dependence structure between commodities and equities as pointed out by previous research, strong comovements at the tails of the predicted distribution could be cause by market-disrupting event, encouraging further parsimony when elaborating on the question of portfolio diversification and restraint when advocating for the diversification property of commodity indices.

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